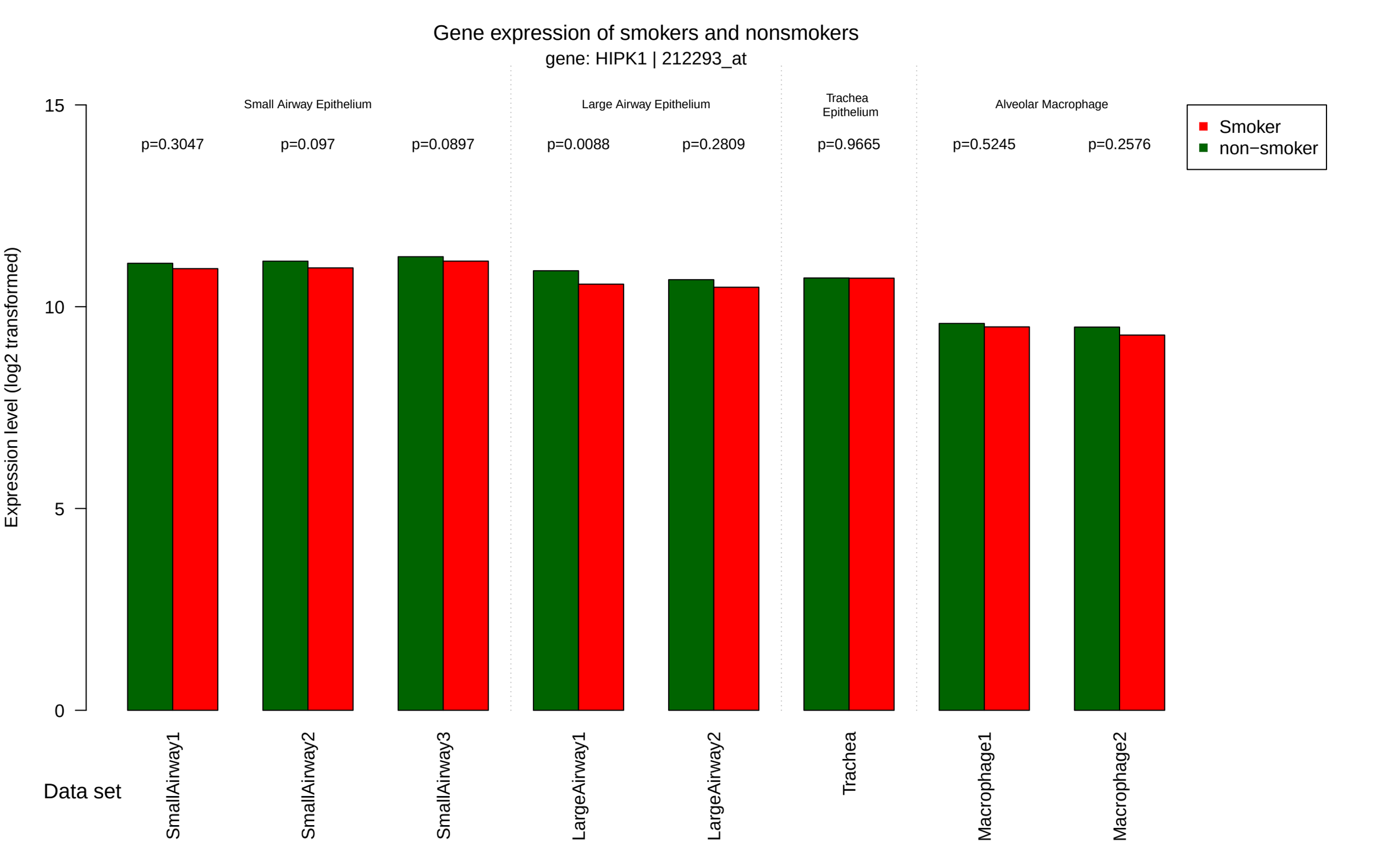
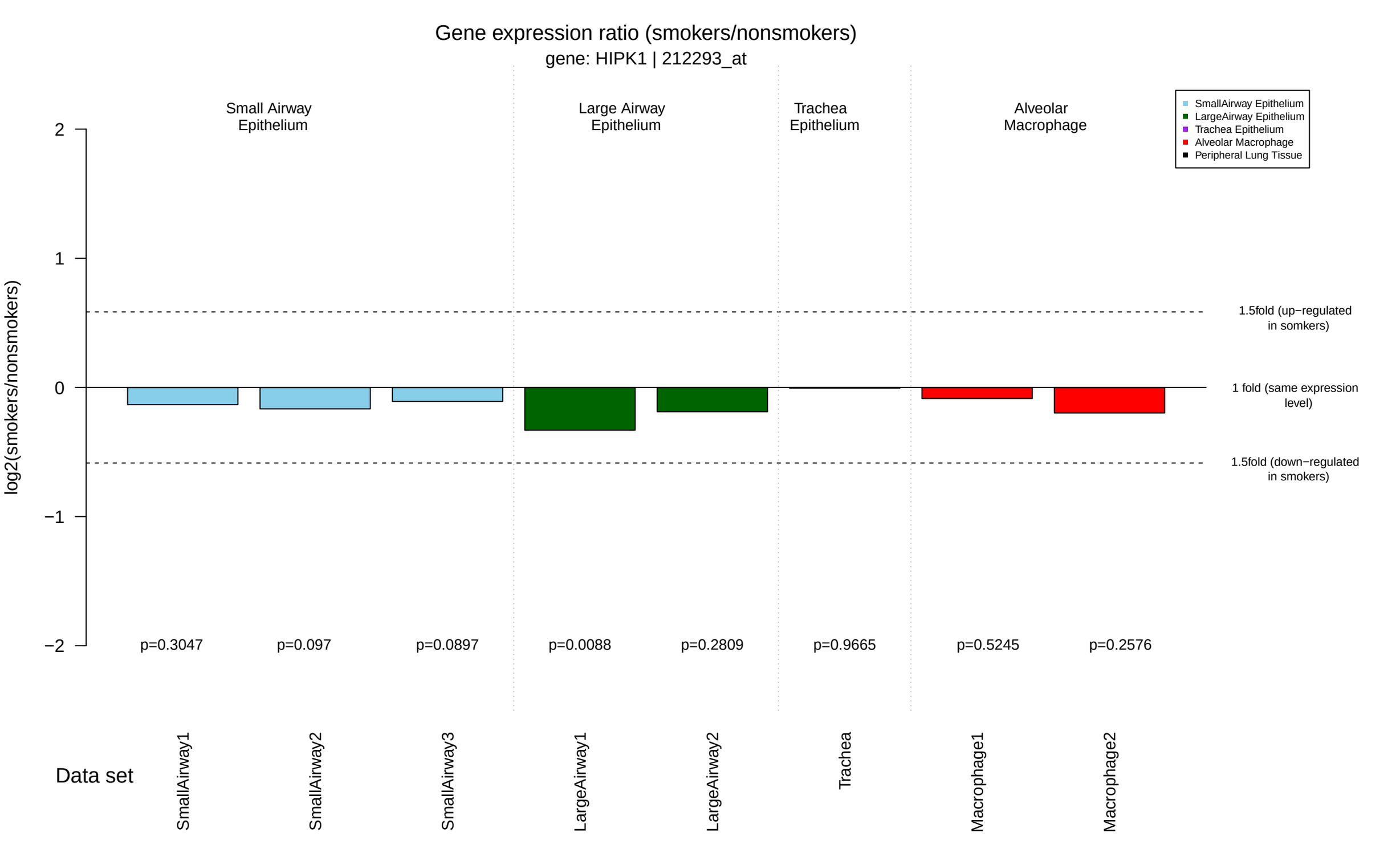
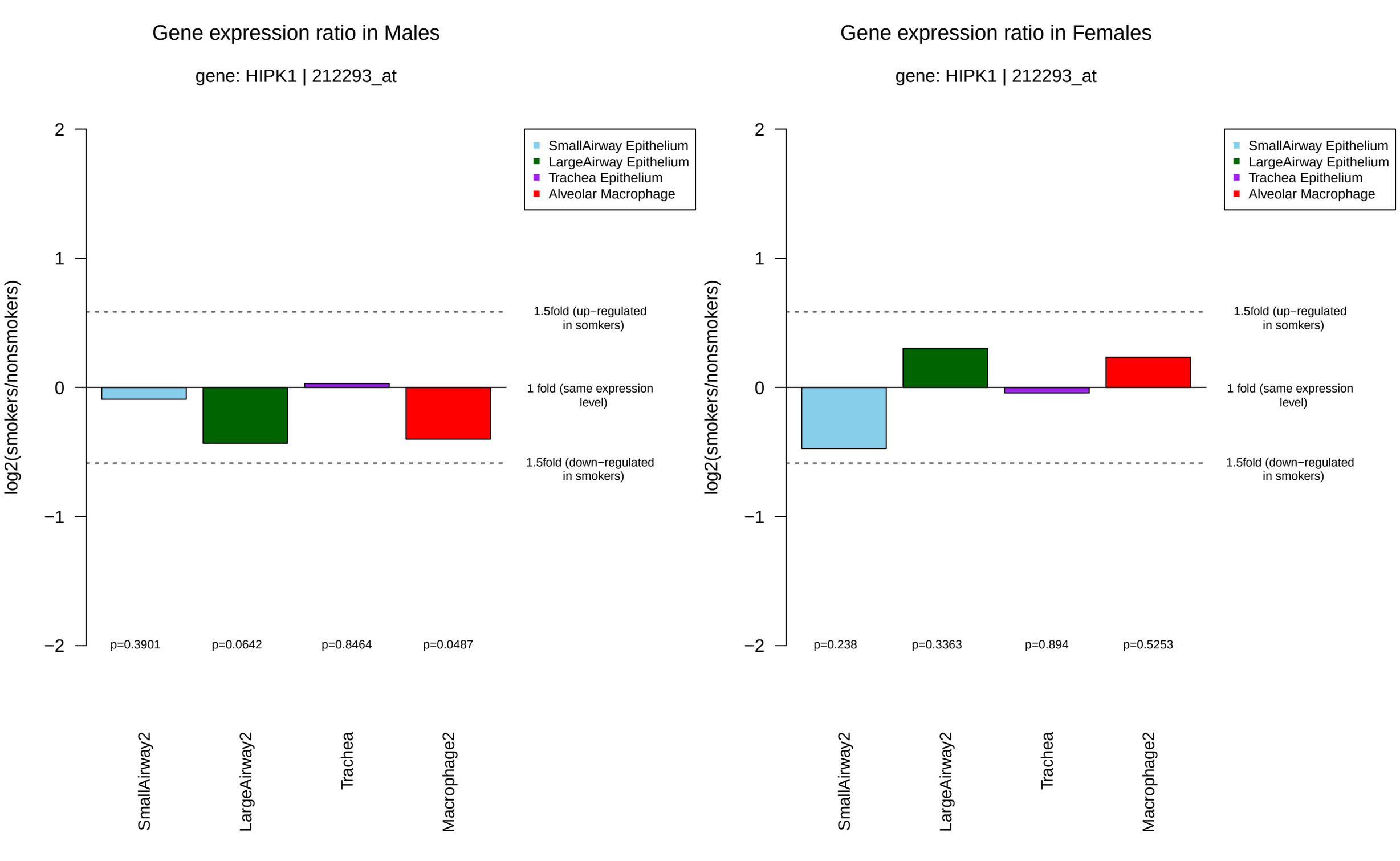
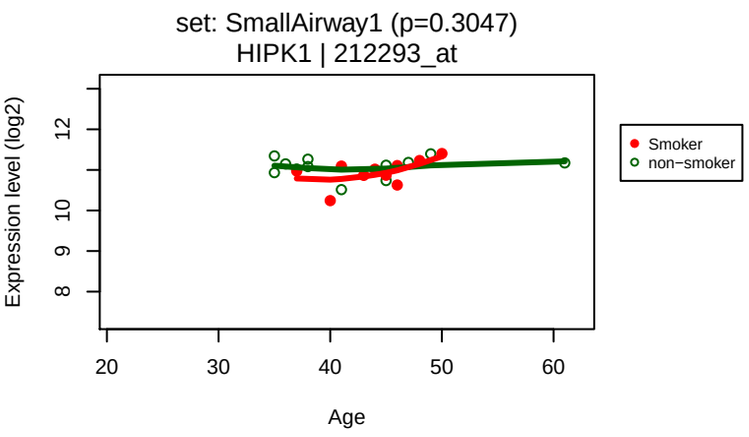
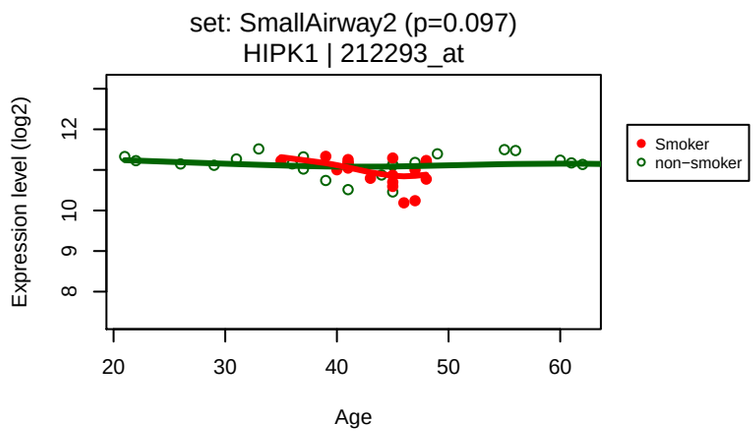
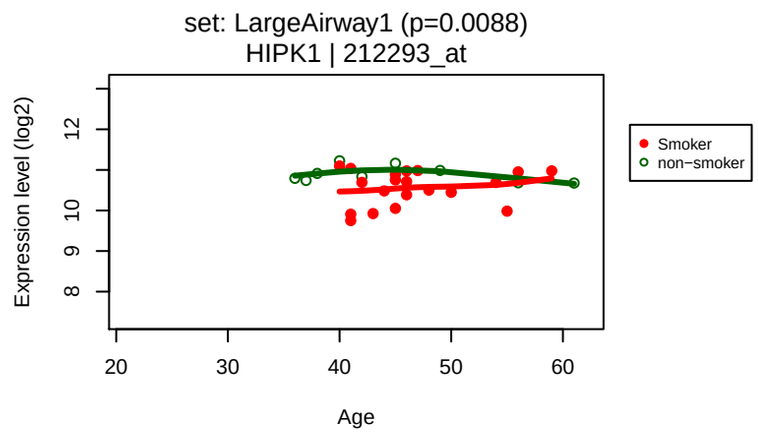

Gene expression of smokers and nonsmokers
gene: HIPK1 | 212293_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3047
p=0.097
p=0.0897
p=0.0088
p=0.2809
p=0.9665
p=0.5245
p=0.2576
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

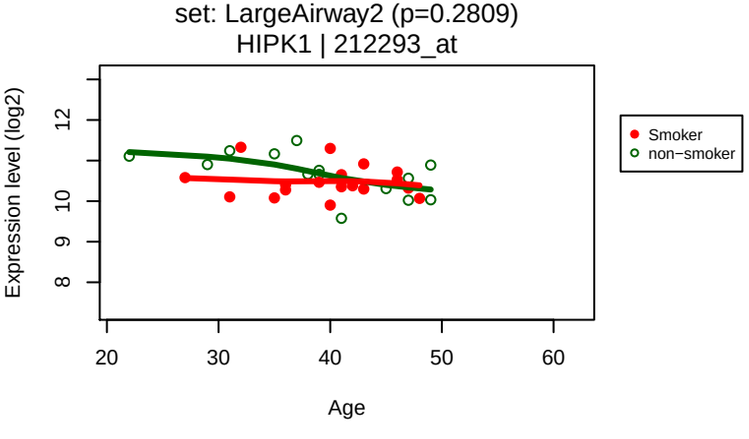

Gene expression ratio (smokers/nonsmokers)
gene: HIPK1 | 212293_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3047
p=0.097
p=0.0897
p=0.0088
p=0.2809
p=0.9665
p=0.5245
p=0.2576
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

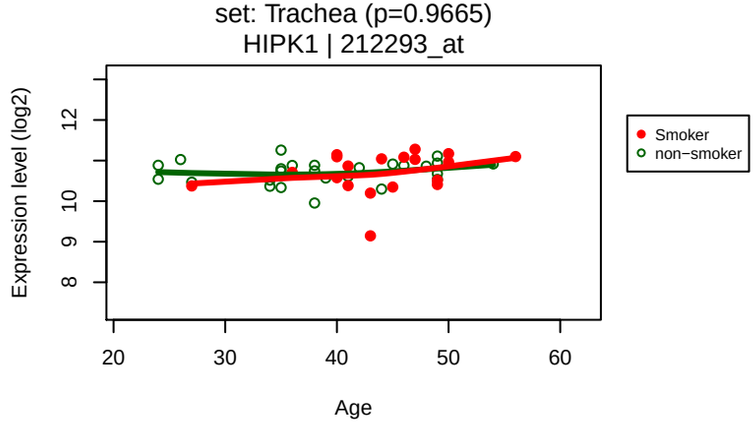

Gene expression ratio in Males
gene: HIPK1 | 212293_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3901
p=0.0642
p=0.8464
p=0.0487
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK1 | 212293_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.238
p=0.3363
p=0.894
p=0.5253
SmallAirway2
LargeAirway2
Trachea
Macrophage2

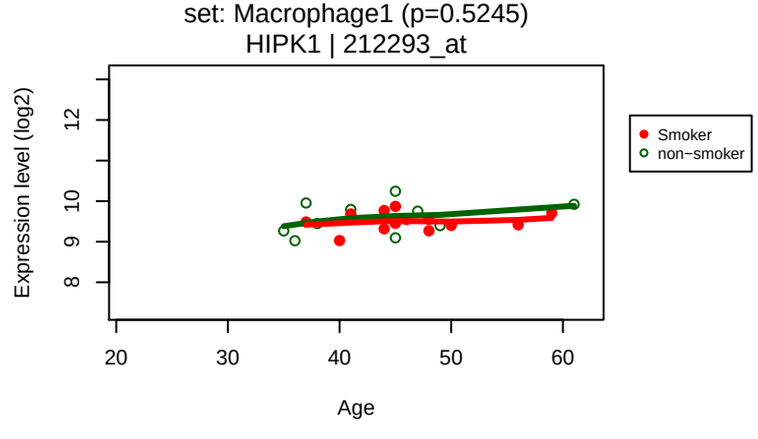

set: SmallAirway1 (p=0.3047)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

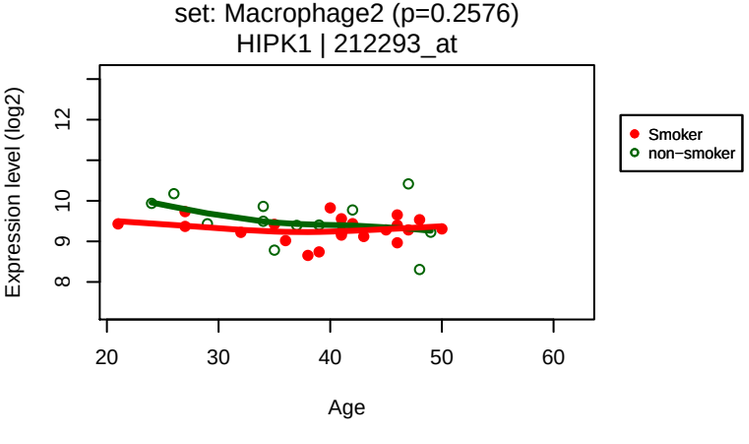

set: SmallAirway2 (p=0.097)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

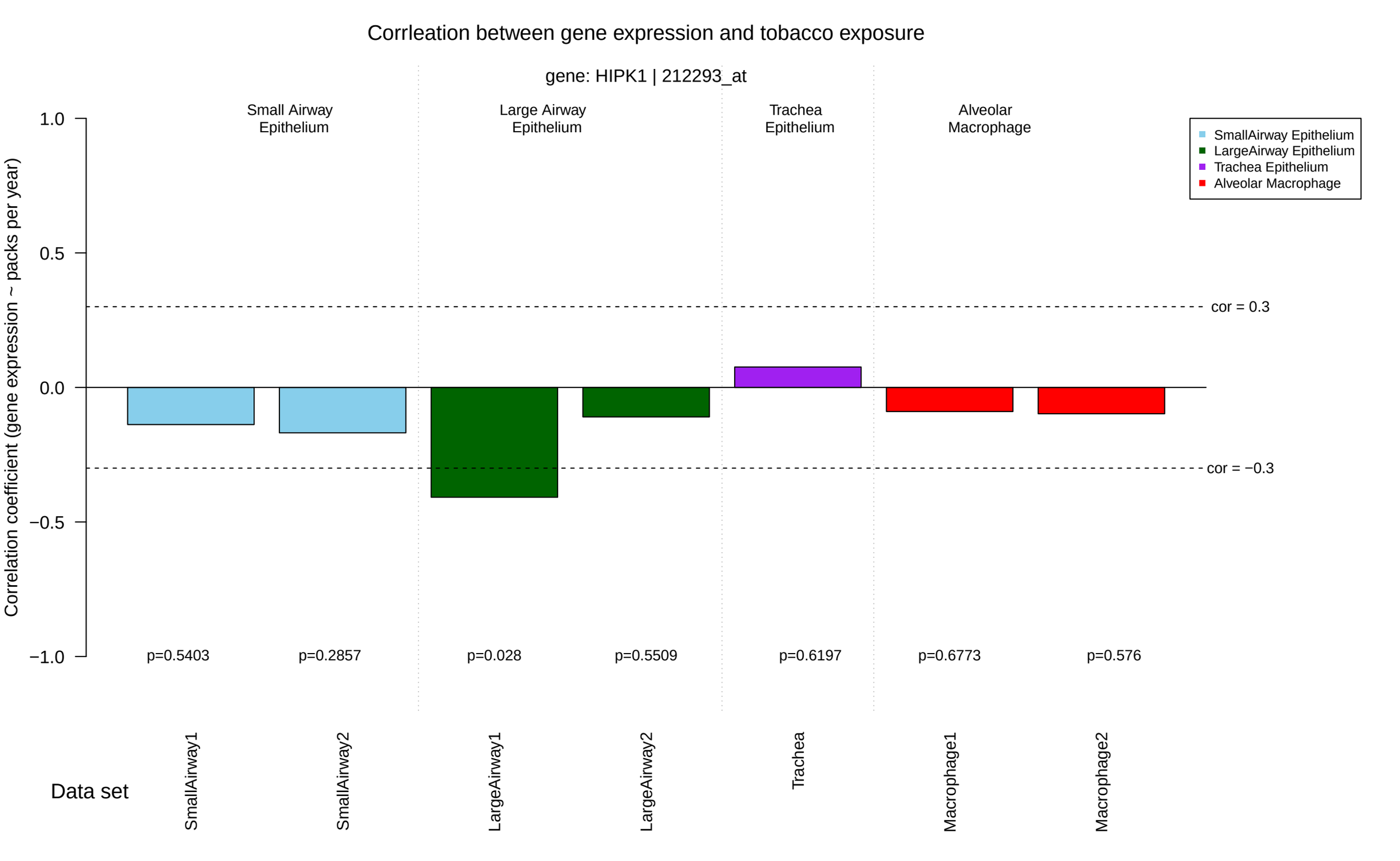

set: LargeAirway1 (p=0.0088)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2809)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.9665)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5245)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2576)
HIPK1 | 212293_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK1 | 212293_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5403
p=0.2857
p=0.028
p=0.5509
p=0.6197
p=0.6773
p=0.576
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2